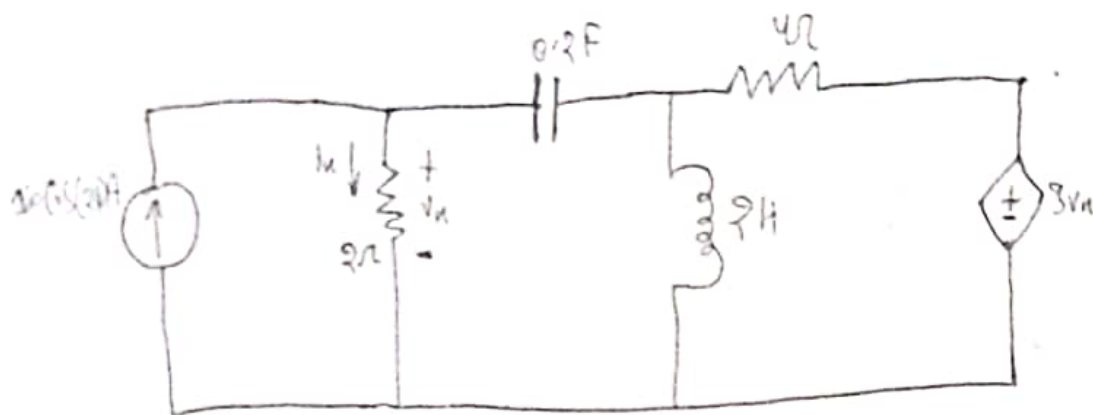


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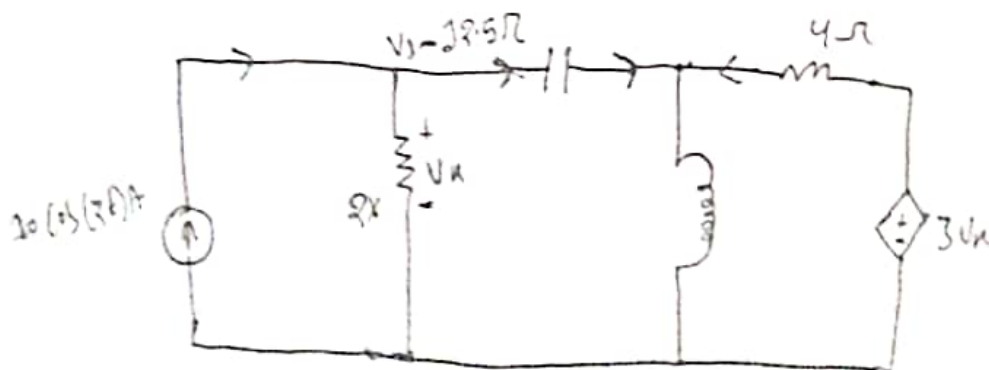
CSE209, Sec-3

Ans to the ques no-1



$$\frac{0.2 F}{Z_C} = \frac{1}{j\omega C} = \frac{1}{j \times 2 \times 0.2} = j2.5 \Omega$$

$$\frac{2 H}{Z_L} = j\omega L = j \times 2 \times 2 = j4 \Omega$$



then, $V_x = V_2$

KCL at node 1,

$$\frac{V_1}{2} + \frac{V_1 - V_2}{-j2.5} = 10$$

$$\Rightarrow V_1(5 + j4) - j4V_2 = 100 \quad \text{--- (1)}$$

KCL at node 2, ...

$$\frac{V_1 - V_2}{-j2.5} + \frac{3V_1 - V_2}{4} = \frac{V_2}{j4}$$

$$\frac{V_1 - V_2}{-j2.5} + \frac{3V_1 - V_2}{4} = \frac{V_2}{j4}$$

$$\Rightarrow V_2(-7.5 + j4) + V_2(2.5 + j1.5) = 0 \quad \text{--- (1)}$$

$$\Delta = \begin{vmatrix} 5 + j4 & -j4 \\ -7.5 + j4 & 2.5 + j1.5 \end{vmatrix}$$

$$= (5 + j4) \times (2.5 + j1.5) - (-7.5 + j4) \times (-j4)$$

$$= 22.5 - j 1.25j$$

$$\Delta_1 = \begin{vmatrix} 100 & -j4 \\ 0 & 2.5 + j1.5 \end{vmatrix}$$

$$= 100 \times (2.5 + j1.5)$$

$$\begin{aligned}
 \therefore V_1 &= \frac{\Delta_1}{\Delta} \\
 &= \frac{100 \times (2.5 \times 21.5)}{22.5 - 21.25} \\
 &= 11.32 \angle 60.01^\circ
 \end{aligned}$$

$$\begin{aligned}
 \Delta'_2 &= \begin{vmatrix} 100 & 2.5 + 21.5 \\ 0 & 7.5 + 24 \end{vmatrix} \\
 &= 100 \times (7.5 \times 24)
 \end{aligned}$$

$$\begin{aligned}
 \therefore V_2 &= \frac{\Delta'_2}{\Delta} \\
 &= \frac{100 \times (7.5 + 24)}{22.5 - 21.25} \\
 &= 33.02 \angle 57.12^\circ
 \end{aligned}$$

$$\therefore V_1(t) = 11.32(0) (2t + 60.01^\circ)^\circ$$

$$V_2(t) = 33.02(0) (2t + 57.12^\circ)^\circ$$